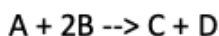


24 Monsanto Plant Disaster. Content warning: #explosion

Tuesday, January 2, 2024
5:53 PM

This problem is taken from Elements of Chemical Reaction Engineering, 5th Edition, by H. Scott Fogler:

A serious accident occurred at the Monsanto plant in Sauget, IL on August 8, 1969 at 12:18am. The blast was heard as far as 10 miles away in Belleville, IL. The explosion occurred in a batch reactor that was used to produce nitroaniline (A) from ammonia (B) and o-nitrochlorobenzene (C):



Where D = NH₄Cl. The reaction was carried out in water solvent (W).

This reaction was normally carried out isothermally at 175°C and about 500 psi. A heat exchanger carrying water at 25°C was used to maintain isothermal operation.

The temperature was controlled by varying the flowrate of water through the heat exchanger. At slower flowrates, the water would get warmer as it flowed through the heat exchanger. At higher flowrates, the water temperature remained constant at 25°C. Maximum cooling hence occurred at higher flowrates.

Isothermal operation was in part a safety feature of this reactor. The reaction is pretty exothermic (heat of reaction = -590 kcal/mol); further it has a pretty low activation energy (11.2 kcal/mol) meaning that the rate increases rapidly with temperature.

Further, at 300°C, nitroaniline decomposes into noncondensable gases such as CO, N₂, and NO₂. Production of noncondensable gases leads to high pressure buildup and hence must be avoided.

The main safety feature on the Monsanto reactor was a rupture disk, which was designed to burst when the pressure in the reactor exceeded 700 psi. Rupture of the disk would decrease the pressure, which would cause the water in the reactor to vaporize, which would remove some of the heat generated by the reaction and hence cool the reactor. Further, removal of steam through the relief valve would remove significant heat from the reactor (Fogler estimates 45,000 kcal/min) and prevent explosion.

Maybe because of this, or maybe for another reason, the reactor was not equipped with a backup cooling system. Instead, when the heat

exchanger would fail, an operator would run out and get it running again. As long as they did this within 10 minutes of heat exchanger failure, there hadn't been any problems.

The day the reactor exploded, several things had gone wrong.

- 1) The reactor was designed for a charge of 3.17 kmol A and 43 kmol B along with 103.7 kmol H₂O. However, the day the reactor exploded, it had been charged with 9.044 kmol A, 33 kmol B, and 103.7 kmol H₂O. It is speculated that Monsanto was interested in increasing production so they increased the charge to the reactor.
- 2) The day the reactor exploded, the heat exchanger failed, and the technician wasn't able to get it working until 10 minutes later.
- 3) The rupture disk also failed.

We are going to model this reaction to show that had any *one* of those things not gone wrong, the reactor wouldn't have exploded.

First, let's model normal reactor operation. The following data and information was taken from Fogler:

- The rate law is $-r_A = kC_A C_B$
 - $k = 0.00017 \text{ m}^3/\text{kmol}\cdot\text{min}$ at 188C
 - $E = 11.3 \text{ kcal/mol}$
 - System volume: 3.26 m³ for "designed" charge, 5.12 m³ for overcharge
 - Heat of reaction = -590 kcal/mol ($= -5.9 \times 10^5 \text{ kcal/kmol}$)
 - $C_{pA} = 0.040 \text{ kcal/mol}\cdot\text{K}$
 - $C_{pW} = 0.018 \text{ kcal/mol}\cdot\text{K}$
 - $C_{pB} = 0.00838 \text{ kcal/mol}\cdot\text{K}$
 - $UA = 35.85 \text{ kcal/min}\cdot\text{K}$
 - The maximum cooling rate is achieved when $T_a = 298 \text{ K}$
- 1) Determine an expression for the maximum cooling rate, $Q_{r,\max}$ for this reactor.
 - 2) Develop the mole balance for isothermal operation. Use it to determine X and Q_g after 45 minutes of operation, which is when the heat exchanger failed. Do this for the intended charge of A as well as the overcharge of A. Compare with the maximum value of Q_r .
 - 3) Now, model adiabatic operation for 10 minutes. What are X , T , and

Q_g for the original and new charge now? Compare again with the maximum value of Q_r .

- 4) Vary the following parameters in order to develop a set of safety criteria that would ensure the reactor would never explode. (We will assume that at least one backup heat exchanger was added to this reactor, that the rupture disk will be tested regularly, and that multiple types of rupture disks have been added. Let's still design for the possibility that everything fails at once.)
- Vary the initial charges of A, B, W assuming the heat exchanger fails for 10 minutes after 45 minutes of operation
 - Vary the time at which the heat exchanger fails, assuming it fails for 10 minutes and that the reactor is charged as designed
 - Vary the amount of time that the heat exchanger is down for, assuming it fails after 45 minutes and that the reactor is charged as designed
 - Vary the time at which the heat exchanger fails, assuming it fails for 10 minutes and that the reactor is overcharged
 - Vary the amount of time that the heat exchanger is down for, assuming it fails after 45 minutes and that the reactor is overcharged

What is the maximum charge to the reactor that ensures an explosion will *never* happen?

$$\textcircled{1} Q_{r, \max} = UA(T - T_a), \text{ where}$$

$$T_a = 298 \text{ K}$$

② mole balance

$$\frac{dX}{dt} = \frac{-r_A}{C_{A0}}$$

$$-r_A = k C_A C_B$$

$$k = k_1 \exp \left(\frac{E}{R} \left(\frac{1}{T_1} - \frac{1}{T} \right) \right)$$

↑
461 K

$$C_A = C_{A0} (1 - X)$$

$$C_B = C_{A0} (\theta_B - 2X) \quad \theta_B = \frac{N_{B0}}{N_{A0}}$$

$$C_{A0} = \frac{N_{A0}}{V}$$

$$V = 3.26 \text{ m}^3 \text{ for designed charge}$$

$$\approx 5.12 \text{ m}^3 \text{ for the overcharge}$$

$$X = 0.068$$

$$Q_g = (r_A)(V)(\Delta H_{rxn})$$

$$\left(-0.0014 \frac{\text{kmol}}{\text{m}^3 \cdot \text{min}} \right) (3.26 \text{ m}^3) \uparrow$$

$$\left(-5.9 \times 10^5 \frac{\text{kcal}}{\text{kmol}} \right) = 2710 \frac{\text{kcal}}{\text{min}}$$

$$Q_{r, \max} = \left(35.85 \frac{\text{kcal}}{\text{min} \cdot \text{K}} \right) (175 + 273 - 278)$$

$$Q_{r, \max} = 5378 \frac{\text{kcal}}{\text{min}}$$

$Q_{r, \max} > Q_g$ i.e. the heat exchanger can control the heat @ this time t in this scenario.

$$\leftarrow 590 \frac{\text{kcal}}{\text{mol}}$$

$$\textcircled{3} \quad T = T_0 + \frac{(-\Delta H_{rxn})(X - 0.068)}{\sum \theta_i C_{pi}}$$

$$= (175 + 273) \text{ K}$$

$\theta_A = 1$
 $\theta_B = \frac{13.06 \frac{\text{kmol}}{\text{m}^3}}{0.906 \frac{\text{kmol}}{\text{m}^3}}$

these values are from the end ($t = 45 \text{ min}$) of isothermal operation, and hence they are the initial values of adiabatic operation.

$$\theta_w = \frac{\left(\frac{103.7 \text{ kmol}}{3.26 \text{ m}^3} \right)}{0.906 \frac{\text{kmol}}{\text{m}^3}}$$

$$X = 0.086, \quad T = 462 \text{ K}, \quad Q_g = 3856 \frac{\text{kcal}}{\text{min}}$$

$$Q_{r, \max} = \left(35.85 \frac{\text{kcal}}{\text{min} \cdot \text{K}} \right) (462 - 298) \text{ K}$$

$$= 5879 \frac{\text{kcal}}{\text{min}}$$

$Q_{r, \max} > Q_g$ so the reaction

won't runaway under this scenario